

Self-Adaptive Model-Based Protection (SAMBPhD)

Complete PhD Research Survey

Full Filesystem Audit · Chronological Inventory · Future Directions

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Abstract

This document provides a **complete, filesystem-verified** chronological survey of all technical work in the SAMBPhD PhD thesis project. The previous survey was incomplete, having been built from session memory rather than a live directory scan. This version is based on a full traversal of `/root/phd_thesis`, including all 55 technical reports (TR-3 through TR-55 for the SG/IBR SyncOC/87L/87B/Distance/Microgrid/Digital-Twin/ML chain), four confirmed IEEE journal papers, six internal SyncOC sub-reports, four additional IBR reports (TR-56, TR-57, TR-58, TR-62), the thesis submission skeleton, and the new_thesis chapter drafts. Git tracking status is noted for every major item.

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1. Inventory Overview

Table 1: Summary of all work items found in `/root/phd_thesis`

Category	Contents	Count
SMBP Technical Reports	TR-3 through TR-62 (with gaps at 59–61, 63–65)	59 reports
SyncOC sub-reports	Internal milestones (report, report2–6)	6 reports
IEEE Journal Papers	6th-order, HIF, coord-gen, line-diff, seq-estim, hif-paper	6 papers
Journal Drafts	Paper A (SyncOC), Paper B (cascade), Paper C (unified)	3 papers
Thesis Skeletons	thesis_submission/, new_thesis/, new_thesis_edit/	3 drafts
Chapter Script Sets	ch4/ (IPR), ch6/ (AI/ML), Z_Final_Thesis_*	3 sets
Simulation Libraries	Z_Synthetic_Family_Library (v1, v2)	2 sets
Presentations	TSA meeting, slides 3–16	15+ files
Total .tex files		≈ 100
Total .py files		≈ 200

Git tracking status: TR-3 to TR-15 and TR-38–45, TR-56–58, TR-62 are fully committed. TR-16–37 and TR-46–55 exist on disk but are **untracked** (never committed to git).

2. Phase 0 — Pre-SAMBPhD: Overleaf Thesis Drafts

Location: `thesis_submission/`, `new_thesis/`, `new_thesis_edit/`, `2-mainmatter/`, `sambp_overleaf/`
Origin: Overleaf syncs (commits: `b863b017`, `ccb693b1`, `5f00135d`, multiple “Update on Overleaf” commits)

The thesis was originally written in Overleaf and synced to this git repository. Three overlapping drafts survive:

`thesis_submission/` (IIT Madras template, most complete)

Uses `iitmdissertation.cls` with compiled `thesis_main.pdf`. Chapter structure:

- Ch. 1: Introduction
- Ch. 2: Modelling (Park dq , reduced-order SG model)
- Ch. 3: Challenges (IBR integration, protection gaps)
- Ch. 4: Inverse Parameter Relay (IPR) — core SyncOC contribution
- Ch. 5: Microgrid Protection
- Ch. 6: AI/ML for Protection
- Ch. 7: Performance Analysis

`new_thesis/` and `new_thesis_edit/` (extended 13-chapter draft)

Chapters cover: Introduction, Mathematical Foundations, IBR Characterisation, Line Differential (87L), Generator OC (SyncOC), Transformer Differential (87T), Distance Protection, Microgrid Protection, Digital Twin, ML/PINN, Centralised Protection, Cybersecurity, Conclusions.

This represents the most ambitious thesis scope — later rationalised into the 6-chapter SAMBPhD structure via `THESIS_CHAPTER_MAP.tex`.

`2-mainmatter/` (earlier chapter-by-chapter drafts)

Raw chapter files: `chap-1Introduction.tex` through `chap-7Performance.tex`, plus `c4_1_sch_adaptive_logic` (the first TikZ circuit schematic committed to git).

`Z_Final_Thesis_*` (data, figures, scripts)

Three parallel directories contain the supporting simulation data and Python scripts for ch4 (IPR), ch5 (Microgrid), and ch6 (AI/ML). Also: `Z_Final_Thesis_data/ch6_generator_aware_selector_v3/` — suggests at least three iterations of the generator-aware model selector were developed.

3. Phase 1 — SyncOC Internal Reports (TR-01 to TR-02B)

Location: `sambp/sambp_sync_oc/report/` through `report6/`
Status: Committed (as part of `sambp_sync_oc` module)

Before the numbered TR series, the SyncOC protection function was developed through a sequence of internal milestone reports:

These led directly to **Paper A** (single-generator SyncOC) and the branching into the numbered TR series.

Table 2: SyncOC internal sub-reports

Report	Title / Topic
report (TR-01)	Inverse-Estimation-Based Adaptive Overcurrent Protection for Synchronous Generators — the foundational LM estimator for SyncOC
report2 (TR-02)	Extended Adaptive Overcurrent Protection Framework for Synchronous Generators — multi-mode extension
report3 (TR-02B)	SAMBPhD-SyncOC HIF Adaptation — high-impedance fault variant
report4	Sixth-Order Park dq Forward Model for SAMBPhD-SyncOC — truth model derivation
report5	(Stage 2 analysis, topic from commit context)
report6	(Integration milestone)

4. Phase 2 — Core SAMBPhD Framework: TR-3 to TR-11

Location: sambp/report3/ through sambp/report11/
Git status: TR-3 to TR-11 all committed (commits 4bb60c40 through 081b3098)

This phase established the SAMBPhD framework across four protection functions and validated it statistically.

TR	Title / Topic	Pages
TR-03	SAMBPhD Model-Based Differential Protection: Transformer 87T and Line 87L using Inverse Estimation and Confidence Gating. First report to cover both 87T and 87L functions jointly. Stage-2 model-veto gate introduced.	~60
TR-04	87B Bus Differential Protection: Extending the SAMBPhD framework to busbar differential. 3-parameter zone model, CT-saturation physics constraint.	~35
TR-05	System Integration Study: OC, 87T, 87L, 87B operating together. 10/10 scenarios selective — first combined pass.	~34
TR-06	IEC 61850 GOOSE Integration: SAMBPhD state sharing over GOOSE for inter-relay coordination.	~32
TR-07	Double-Busbar Coordination: Bus-coupler check zone added to 87B. Complex topology with bus-section switches.	~29
TR-08	Monte Carlo Robustness Study: 4000 trials, 20 scenarios. First statistical validation of the framework.	~42
TR-09	Stage-2 Confidence Gate Recalibration and Extended MC: TPR = 1.000, FPR = 0.000 achieved after recalibration.	~35
TR-10	CT Saturation Injection Study: Synthetic CT saturation waveforms injected; Stage-2 veto gate validated.	~32
TR-11	High-Impedance Fault Sensitivity Margin Study: HIF detection threshold sensitivity under varying soil resistivity.	~29

Key result from this phase: The Physics-Reduction Proposition formally established — $\kappa_n < 80$ across all four SG protection functions. Stage-2 model-veto gate achieves TPR = 1.00, FPR = 0.00 at 4000 trials.

5. Phase 3 — 87L Deep Dive: TR-12 to TR-27

Location: sambp/report12/ through sambp/report27/

Git status: TR-12 to TR-15 committed (c6adb8be, a916e87b, f783389a); TR-16 to TR-27 untracked on filesystem

A deep, systematic investigation of the 87L line differential function under complex scenarios.

TR	Title / Topic	Lines
TR-12	87L Magnitude Override: Alternative restraint criterion when LM solution is ill-conditioned ($\kappa_n > \text{threshold}$).	370
TR-13	87L Pi-Section Charging Compensation: Long-line capacitive charging current subtracted before differential computation.	532
TR-14	Dynamic Busbar Transfer with GOOSE Zone Re-configuration: 87L zone boundaries updated dynamically via IEC 61850 GOOSE.	480
TR-15	87L Voltage-Based Pi-Section Correction: Improved charging correction using measured terminal voltages.	473
TR-16	87B/87L Coordination Study: 28/28 scenarios all selective with coordinated 87B and 87L.	425
TR-17	Adaptive 87L Threshold: Dynamic restraint threshold as a function of I_{rest} .	370
TR-18	Mho Distance Relay (21/21P): First distance relay introduction, coordinated with 87B/87L.	389
TR-19	Simultaneous Multi-Feeder Busbar Transfer: Coordination during concurrent busbar switching events.	344
TR-20	Frequency-Adaptive 87L Charging-Current Correction: Off-nominal frequency compensation for micro-grid applications.	450
TR-21	Harmonic-Component Discrimination for 87L: 2nd/5th harmonic blocking for inrush current in meshed networks.	386
TR-22	Negative-Sequence 87LN Line Differential: Asymmetric fault (SLG, LLG) detection via negative-sequence component.	413
TR-23	Transient Differential Current Supervision (TDCS-87L): Transient blocking for 3PH close-in faults in IBR-rich networks.	386
TR-24	Adaptive TDCS Threshold: Variable TDCS threshold for changing-baseline microgrid networks.	377
TR-25	Zero-Sequence 87LN0 Line Differential: Detection of high-resistance earth faults in pure-IBR networks lacking zero-sequence current.	341
TR-26	IBR Parametric Monte Carlo Robustness Study: 15k trials across IBR penetration levels; confirmed selectivity with IBR fault current.	334

TR	Title / Topic	Lines
TR-27	Updated System Integration: IBR-Aware 87L Chain: Updated multi-function study incorporating all 87L improvements from TR-12–26.	325

6. Phase 4 — Synthesis and Distance Protection: TR-28 to TR-37

Location: sambp/report28/ through sambp/report37/
Git status: All **untracked** (exist on filesystem, never committed)

TR	Title / Topic	Lines
TR-28	Master Synthesis TR-03 through TR-27: Unified results table, gap analysis, and evidence map for all work up to this point.	500
TR-29	IBR Apparent Impedance Trajectory Analysis for Distance Relay 21: How IBR current-limiting shifts apparent impedance into mho characteristic — leading to false trips.	495
TR-30	Cross-Polarised Mho Characteristic for IBR Networks: Polarisation using memorised positive-sequence voltage avoids false-trip during LVRT.	418
TR-31	Zero-Sequence Compensation for IBR Earth Faults (21G): k_0 compensation factor re-derived for grids with IBR zero-sequence injection.	410
TR-32	Negative- and Zero-Sequence Directional Elements (67Q, 67N): IBR-aware directional elements for asymmetric faults on IBR-fed lines.	375
TR-33	Permissive Overreaching Transfer Trip (POTT) for IBR Lines: Communication-aided tripping scheme; POTT logic validated against IBR impedance trajectories.	388
TR-34	Quadrilateral Distance Characteristic for IBR Lines: Quadrilateral replaces circular mho for improved reach setting under high-resistance IBR faults.	359
TR-35	Full IBR Distance Protection Coordination Study: Combined 21/67Q/67N/POTT in a test network with 40% IBR penetration.	376
TR-36	Auto-Reclosing Strategy for IBR Lines: Dead-time window adjusted for IBR recovery transient; prevents re-trip on restoration current.	319
TR-37	Monte Carlo Reliability Assessment of IBR Protection Scheme: End-to-end statistical validation of the distance protection chain.	300

7. Phase 5 — Microgrid Protection: TR-38 to TR-42

Location: sambp/report38/ through sambp/report42/, sambp/sambp_system/
Git status: TR-38 to TR-42 committed (3ae4c5cf)

TR	Title / Topic	Lines
TR-38	GFL vs. GFM Inverter Fault Characterisation: Systematic comparison of fault current profiles — GFL current-limited vs. GFM synchronising.	353
TR-39	Islanding Detection and Protection Mode Switching: ROCOF + voltage-vector-shift algorithm; protection mode handover between grid-connected and islanded.	285
TR-40	Adaptive Overcurrent Coordination for IBR Microgrids: OC relay coordination re-computed dynamically when microgrid topology changes.	266
TR-41	87B Bus Differential in IBR Microgrids: 87B behaviour under IBR fault current (low magnitude, fast decay).	262
TR-42	Microgrid Protection System Integration Study: Combined test of all microgrid protection elements — selectivity, speed, security.	285

Supporting code: sambp/sambp_system/ contains a comprehensive set of study runners including: run_gfl_gfm_study.py, run_islanding_study.py, run_adaptive_oc_study.py, run_87b_mg_study.py, run_mg_integration_study.py, run_ar_study.py (auto-reclose), and more.

8. Phase 6 — Digital Twin: TR-43 to TR-45

Location: sambp/report43/ through sambp/report45/
Git status: Committed (a93ab551)

TR	Title / Topic	Lines
TR-43	Digital Twin Architecture for IBR Protection Systems: Real-time shadow model of the protection system running in parallel with hardware relay.	261
TR-44	Real-Time Parameter Estimation and State Tracking (EKF): Extended Kalman Filter for multi-parameter estimation in the digital twin.	261
TR-45	Digital Twin Validation and Protection System Integration: DT-relay agreement rate as a new performance metric; validation framework.	252

9. Phase 7 — ML/PINN and Advanced Topics: TR-46 to TR-55

Location: sambp/report46/ through sambp/report55/
Git status: All **untracked** on filesystem

TR	Title / Topic	Lines
TR-46	Physics-Informed Neural Network (PINN) for IBR Fault Classification: PINN architecture embedding physics constraints (KCL, KVL) as loss terms.	230
TR-47	Feature Engineering and Physics Constraints for ML Protection: Feature set definition; constraint satisfaction layer in the network architecture.	253
TR-48	Online Learning and Adaptive ML for Changing IBR Fleets: Continual learning with drift detection for evolving IBR penetration.	259
TR-49	ML Protection System Integration and Validation: Full pipeline from raw waveform to trip decision via PINN.	261
TR-50	Wide-Area Protection Coordinator (WAPC): Centralised coordinator receiving synchrophasor data from all relays; system-level consistency check.	313
TR-51	Automatic Protection Settings Optimisation: Relay settings optimiser under protection selectivity and grading constraints.	306
TR-52	Cybersecurity of IEC 61850 GOOSE Networks: Threat modelling and intrusion detection for GOOSE-based relay coordination.	296
TR-53	Hardware-in-the-Loop Validation Protocol: HIL test harness architecture; RTDS interface specification.	271
TR-54	Field Deployment and Commissioning Guide: Step-by-step commissioning workflow for the full SAMPB system.	285
TR-55	Thesis Synthesis and Contribution Map: Report inventory and chapter mapping — the pre-IBR synthesis of all TR-3 to TR-54 work.	306

Supporting code: `sampb/sampb_system/` contains matching runners: `run_pinn_study.py`, `run_feature_engineering_study.py`, `run_online_learning_study.py`, `run_wapc_study.py`, `run_settings_optimisation_study.py`, `run_cybersecurity_study.py`, `run_hil_validation_study.py`, `run_deployment_study.py`, `run_thesis_synthesis.py`.

10. Phase 8 — IEEE Journal Papers (Parallel Track)

Locations: `ieee_6th_order/`, `ieee_hif_tpwd/`, `ieee_coord_gen/`, `ieee_line_diff/`, `ieee_seq_estim/`, `sampb/hif_paper/`
Git status: All committed (various commits, latest `a1b80fa5`, `0c31b6e1`)

Six IEEE-format journal papers were developed in parallel with the TR series:

Presentation materials: `TSA_meeting_Eluvathingal.tex` (TSA committee meeting slides), `slide03` through `slide16` (16 study-note slides for individual TR summaries).

11. Phase 9 — IBR Extension: TR-56, TR-57, TR-58, TR-62

Location: `sampb/report56/`, `report57/`, `report58/`, `report62/`
Git status: TR-57, TR-62 committed; TR-56, TR-58 untracked (content complete, PDF built)

Table 9: IEEE journal papers

Folder	Title	Target
ieee_6th_order	Validation Boundaries of Reduced-Order Inverse Protection Models: 6th-order Park dq truth model; $L^* \approx 150$ ms validity boundary; AVR as primary driver	IEEE
ieee_hif_tpwd	Impedance-Aware Adaptive Very-Inverse Overcurrent Protection for High-Impedance Faults in Synchronous-Generator Feeders	IEEE TPWRD
sampb/hif_paper	(Same HIF paper, earlier draft with <code>hif_sim.py</code> simulation)	IEEE TPWRD
ieee_seq_estim	Sequence-Component Inverse Estimation for Adaptive Overcurrent Protection	IEEE
ieee_coord_gen	Coordinated Adaptive Overcurrent Protection for Parallel Generator Systems	IEEE
ieee_line_diff	IBR-Adaptive Line Differential Protection: Harmonic Discrimination, Pi-Section Compensation, and IBR Fault Classification (8 pages, full journal depth)	IEEE TIE

Motivation: RESEARCH_GAP_PLAN.md identifies five gaps (G1–G5) not covered by TR-3 to TR-55.

11.1 TR-56 — DFIG Piecewise-ODE Model (87L/87G)

Chapter: Ch. 6 §6.1 **Target:** IEEE TEC 715 lines

Problem: DFIG fault current has two regimes (RSC-active then crowbar-fired). SG 4-parameter model produces false-block rate $> 30\%$.

10-parameter model: $\theta = [k_{ibr}, \varphi_{pre}, t_{cb}, I_{fund}, \varphi_{fund}, I_{nat}, \varphi_{nat}, \omega_{slip}, \tau_s, I_{DC}]$

Physics collinearity reduction: $\tau_{DC} = \tau_s$ (DC and natural-response share stator transient time constant). Smoothed Heaviside ($\sigma_h = 0.5$ ms) for LM-compatible crowbar transition. CUSUM crowbar detector (2-cycle windows, 2048-point FFT): detection latency 10 ms.

Table 10: κ_n vs. slip — all within threshold

Slip s	ω_{slip} [rad/s]	κ_n	Accept (< 80)?
3%	9.4	67	✓
10%	31.4	30	✓
20%	62.8	19	✓
35%	110.0	13	✓

11.2 TR-57 — Bayesian 3-Hypothesis SPRT (87T IBR)

Chapter: Ch. 4 **Target:** IEEE TPWRD 934 lines

Per-half-cycle asymmetry index A_k drives a three-hypothesis SPRT (H_0 : Inrush, H_1 : IBR fault, H_2 : CT saturation). Wald boundaries $\log A = \pm 6.91$ ($\alpha = \beta = 0.001$). Trip time 20 ms vs. ≥ 60 ms IEC 60255-111.

11.3 TR-58 — Generalised EAC + CUEP Bound (Relay 78)

Chapter: Ch. 6 §6.3 **Target:** IEEE TPWRD 892 lines

Conservative CUEP lower bound (Proposition 1): $P_{max}^{eff} = P_{max}^{SG} - \Delta P_{max}^{GFL}$, $\delta_{LB} = \pi - \arcsin(P_m/P_{max}^{eff}) \leq \delta_u$ (proven). Conservatism gap $\varepsilon \leq 7.2^\circ$. Standard blinder error: up to $+6.7^\circ$ (unsafe).

11.4 TR-62 — 2-Parameter PV Zone Model (87L)

Chapter: Ch. 3 **Target:** IEEE TIE 615 lines

PV SRF controller drives DC to zero in ≤ 15 ms: $\theta_{PV} = [I_{fund}, \varphi_{fund}]$, $\kappa_n = 1.00$ analytically ($36\times$ improvement over SG model). DDR-based model selector separates PV from SG.

12. Phase 10 — SAMPB Architecture Lock-in and Paper C

Locations: THESIS_CHAPTER_MAP.tex, sambp/sambp_sync_oc/paper_c/, sambp/sambp_sync_oc/paper/, sambp/sambp_sync_oc/paper_b/
Git status: Committed

Paper A — Real-Time Inverse Parameter Estimation for Adaptive OC Protection

8 pages, full journal depth. Target: *IEEE Transactions on Power Delivery*. Single-generator SyncOC, composite confidence gate, multi-mode transition.

Paper B — Hilbert–Fortescue Sequence Estimation and Greedy Cascade Coordination

Target: *IEEE Transactions on Industrial Electronics*. Multi-generator cascade (M4), Hilbert–Fortescue symmetrical-component extension (M3).

THESIS_CHAPTER_MAP.tex

Locked TR-to-chapter allocation across 6 technical chapters (Ch. 2–7). 5-rule new-TR checklist. Critical path established: TR-56 $\rightarrow$ TR-65 $\rightarrow$ TR-67 $\rightarrow$ Q3‘28.

Paper C (PaperF) — Unified SAMPB Framework

7 pages after Phase 2 update (a4222f73). Phase 1: OC, 87T, 87L, 87B for SG networks (26/28 CCR, ≤ 20 ms). Phase 2 (early 2027): adds DFIG/PV/Relay-78/gen-suite (≥ 62 scenarios). Target: *IEEE Transactions on Power Systems*.

13. Phase 11 — Additional SMBP Modules

Locations: sambp/sambp_distance/, sambp/sambp_double_bus/, sambp/sambp_coordination/, sambp/sambp_generator/, sambp/diagrams_tikz/, sambp/sambp_diagrams/

Table 11: SMBP Python module directories

Module	Contents
sambp_sync_oc/	SyncOC: models, inverse_estimation, adaptation, evaluation, signal_processing, config
sambp_line_diff/	87L: line zone models (SG 4-param, PV 2-param, DFIG 10-param), DDR selector, estimators
sambp_transformer_diff/	87T: Jiles–Atherton inrush model, SPRT, confidence gate
sambp_bus_diff/	87B: 3-param CT-saturation model, Stage-2 veto
sambp_generator/	Generator suite: DFIG model (TR-56), CUSUM detector, 2-pass LM estimator
sambp_distance/	Distance relay (21/21P): mho, quadrilateral, cross-polarisation
sambp_double_bus/	Double-busbar: bus-coupler check zone, GOOSE zone reconfiguration
sambp_coordination/	87B/87L coordination, simultaneous transfer studies
sambp_system/	Full system coordinator: 30+ study runners covering all TR-38 to TR-55 topics
diagrams_tikz/	TikZ circuit schematics (common to all reports)

14. Supporting Infrastructure

Table 12: Supporting directories and their role

Directory	Role
Z_Synthetic_Family_Library/	Synthetic fault waveform library v1: waveforms, metadata, benchmark results
Z_Synthetic_Family_Library_v2/	v2 with benchmark_results_v4 subdirectory (extended parameter sweep)
PDF_COLLECTION/	Organised PDF archive: 01 SMBP Reports, 02 IEEE Papers, 03 Thesis, 04 Chapter Docs
FIGURE_ARCHIVE/	Rendered PNGs (F1) and LaTeX source (F2) for all figures
ch4/	Chapter 4 simulation scripts: plotting/, simulation/
ch6/	Chapter 6 scripts: metrics/, plotting/, protection/, schematics/, selector/, simulation/, training/, validation/
data/	Processed data: chapter_4/, chapter_6/, simulations/, validation/
analytics_validation/	Master trainer, batch processor, ICI audit engine, security/latency audits
RESEARCH_GAP_PLAN.md	5-gap research plan that launched TR-56 through TR-67
THESIS_CHAPTER_MAP.tex	Locked TR→chapter mapping
sambp_overleaf/	Overleaf-compatible SMBP figures and TikZ diagrams

15. What the Previous Survey Missed

The first survey (built from session memory) missed:

1. **TR-3 to TR-11 (Phase 2):** The foundational 87T/87L/87B work, 4000-trial Monte Carlo, CT saturation study, HIF sensitivity study. These are the core experimental validation of the Physics-Reduction Proposition.
2. **TR-12 to TR-27 (Phase 3):** All 16 87L deep-dive reports — pi-section compensation, harmonic discrimination, negative-sequence, TDCS, GOOSE zone reconfiguration, IBR 15k-trial Monte Carlo.
3. **TR-28 to TR-37 (Phase 4):** The complete distance protection track — apparent impedance trajectories, cross-polarised mho, zero-sequence compensation, directional elements, POTT, auto-reclose, reliability Monte Carlo.
4. **TR-38 to TR-42 (Phase 5):** Microgrid protection — GFL/GFM characterisation, islanding detection, adaptive OC coordination.
5. **TR-43 to TR-45 (Phase 6):** Digital twin architecture, EKF parameter estimation, DT-relay agreement validation.
6. **TR-46 to TR-55 (Phase 7):** PINN, online ML, WAPC, settings optimisation, cybersecurity, HIL protocol, deployment guide, thesis synthesis.
7. **Four additional IEEE papers:** `ieee_seq_estim` (sequence estimation), `ieee_coord_gen` (parallel OC coordination), `ieee_hif_tpwd` (HIF overcurrent), `sambp/hif_paper`.
8. **SyncOC internal sub-reports:** 6 milestone reports in `sambp_sync_oc/report` through `report6`.
9. **Pre-SAMBPhD thesis drafts:** `thesis_submission/`, `new_thesis/` (13-chapter draft), `2-mainmatter/`, and all `Z_Final_Thesis_*` data/figures/scripts.

16. Complete TR Inventory Table

TR	Phase	Topic	Git
TR-03	SG core	87T + 87L: Differential Protection Framework	✓
TR-04	SG core	87B Bus Differential Protection	✓
TR-05	SG core	System Integration (10/10 selective)	✓
TR-06	SG core	IEC 61850 GOOSE Integration	✓
TR-07	SG core	Double-Busbar Coordination	✓
TR-08	SG core	Monte Carlo Robustness (4000 trials)	✓
TR-09	SG core	Stage-2 Recalibration (TPR=1.00, FPR=0.00)	✓
TR-10	SG core	CT Saturation Injection Study	✓
TR-11	SG core	HIF Sensitivity Margin Study	✓
TR-12	87L deep	87L Magnitude Override	✓
TR-13	87L deep	87L Pi-Section Charging Compensation	✓
TR-14	87L deep	Dynamic Busbar Transfer + GOOSE Zone Re-config	✓
TR-15	87L deep	87L Voltage-Based Pi-Section Correction	✓
TR-16	87L deep	87B/87L Coordination (28/28 scenarios)	untracked
TR-17	87L deep	Adaptive 87L Threshold vs. I_{rest}	untracked
TR-18	87L deep	Mho Distance Relay 21/21P	untracked
TR-19	87L deep	Simultaneous Multi-Feeder Transfer	untracked
TR-20	87L deep	Frequency-Adaptive 87L Charging Correction	untracked
TR-21	87L deep	Harmonic Discrimination for 87L	untracked
TR-22	87L deep	Negative-Sequence 87LN Line Differential	untracked
TR-23	87L deep	TDCS-87L Transient Supervision	untracked
TR-24	87L deep	Adaptive TDCS Threshold	untracked
TR-25	87L deep	Zero-Sequence 87LN0 for Pure-IBR Earth Faults	untracked
TR-26	87L deep	IBR Parametric Monte Carlo (15k trials)	untracked
TR-27	87L deep	Updated IBR-Aware 87L Chain Integration	untracked
TR-28	Synthesis	Master Synthesis TR-03 to TR-27	untracked
TR-29	Distance	IBR Apparent Impedance Trajectory Analysis	untracked
TR-30	Distance	Cross-Polarised Mho for IBR Networks	untracked
TR-31	Distance	Zero-Sequence Compensation for IBR Earth Faults	untracked
TR-32	Distance	Negative/Zero-Sequence Directional Elements	untracked
TR-33	Distance	POTT Scheme for IBR Lines	untracked
TR-34	Distance	Quadrilateral Distance Characteristic	untracked
TR-35	Distance	Full IBR Distance Protection Coordination	untracked
TR-36	Distance	Auto-Reclosing Strategy for IBR Lines	untracked
TR-37	Distance	Monte Carlo Reliability — IBR Distance Scheme	untracked
TR-38	Microgrid	GFL vs. GFM Fault Characterisation	✓
TR-39	Microgrid	Islanding Detection + Mode Switching	✓
TR-40	Microgrid	Adaptive OC Coordination for IBR Microgrids	✓
TR-41	Microgrid	87B in IBR Microgrids	✓
TR-42	Microgrid	Microgrid Protection System Integration	✓
TR-43	DT	Digital Twin Architecture	✓
TR-44	DT	Real-Time EKF Parameter Estimation	✓
TR-45	DT	Digital Twin Validation + Integration	✓
TR-46	ML/PINN	PINN for IBR Fault Classification	untracked
TR-47	ML/PINN	Feature Engineering + Physics Constraints	untracked
TR-48	ML/PINN	Online Learning for Changing IBR Fleets	untracked

TR	Phase	Topic	Git
TR-49	ML/PINN	ML Protection System Integration	untracked
TR-50	Advanced	Wide-Area Protection Coordinator (WAPC)	untracked
TR-51	Advanced	Automatic Settings Optimisation	untracked
TR-52	Advanced	IEC 61850 GOOSE Cybersecurity	untracked
TR-53	Advanced	HIL Validation Protocol	untracked
TR-54	Advanced	Field Deployment & Commissioning Guide	untracked
TR-55	Synthesis	Thesis Synthesis and Contribution Map	untracked
TR-56	IBR gen.	DFIG Piecewise-ODE Model (87L/87G)	✓
TR-57	IBR 87T	Bayesian 3-Hypothesis SPRT (87T IBR)	✓
TR-58	IBR gen.	Generalised EAC + CUEP Bound (Relay 78)	✓
TR-62	IBR 87L	2-Parameter PV Zone Model (87L)	✓

Table 13: Complete TR inventory (59 reports)

17. Future Directions

17.1 Immediate — Q2/Q3 2026

TR-59: Virtual 87G-DFIG via Slip-Frequency EKF (Ch. 6 §6.2, IEEE TEC) — DFIG generator differential without rotor CTs, using TR-56’s slip-frequency observable as EKF state.

TR-60: Relay 40 LOE with GFM IBR Admittance Correction (Ch. 6 §6.4, IEEE TPWRD) — GFM virtual admittance shifts mho locus; corrected LOE mho setting from commissioning admittance parameters.

TR-61: 64G Stator Earth Fault for IBR-Augmented Machine Buses (Ch. 6 §6.5, IEEE TIE) — Optimal sub-harmonic injection frequency f^* re-derived for IBR-modified bus admittance.

17.2 Medium Term — Q4 2026 / Q1 2027

TR-63: Relay 81 IFDI for Zero-Inertia GFM Networks (Ch. 6 §6.6, IEEE TSG) — Dynamic virtual inertia tracking via IFDI; replaces static ROCOF calibration.

TR-64: Complete IBR-Aware 87T Integration + 5000-Trial MC (Ch. 4, IEEE TPWRD) — Journal-ready version of TR-57 with full MC validation.

TR-65: Comprehensive Generator Protection Suite (Ch. 7, IEEE TPWRD) — Unifies TR-56, 58, 59, 60, 61, 63 into one framework paper; supersedes Paper C Phase 2 section.

17.3 Long Term — Q2 2027 / Q1 2028

TR-66: 10,000-trial MC per protection element (cluster compute).

TR-67: RTDS HIL with 2 MW DFIG emulator + PV inverter emulator.

Paper C revised submission (early 2027): ≥ 62 scenarios across all 9 protection functions; Phase 1 HIL results added.

17.4 Open Research Gaps

- Multi-machine EAC extension (TR-58 Proposition 1 is single-machine only; EEAC with coherent decomposition needed for multi-machine substations)
- Asymmetric fault handling for DFIG (TR-56 covers balanced 3PH only; SLG requires sequence-domain decomposition)
- GFM IBR loss of synchronism detection (distinct from SG out-of-step; not in current gap plan)
- Git hygiene: TR-16 to TR-37, TR-46 to TR-55 (29 reports) are untracked and at risk of loss — these should be committed

18. Central Thesis Claim

Claim (to be proven by Q3 2028):

A single physics-parameter-reduction discipline — identifying and analytically deriving the most collinear parameter in each zone model — produces $\kappa_n < 80$ across all nine protection functions (OC, 87T-SG, 87T-IBR, 87L-SG, 87L-PV, 87L-DFIG, 87B, generator suite, Relay 78), enabling $\geq 99\%$ correct classification across 10 000 Monte Carlo scenarios, without function-specific signal-processing heuristics.

Critical evidence gap: Monte Carlo validation for the four IBR functions (TR-56, TR-57/64, TR-62, TR-58) is currently deterministic only. TR-66 closes this gap.

Critical path:

$$\underbrace{\text{TR-56}}_{\checkmark} \rightarrow \underbrace{\text{TR-64/65}}_{\text{Q4'27}} \rightarrow \underbrace{\text{TR-66 MC}}_{\text{Q1'28}} \rightarrow \underbrace{\text{TR-67 HIL}}_{\text{Q1'28}} \rightarrow \underbrace{\text{Paper C rev.}}_{\text{early'27}} \rightarrow \underbrace{\text{Thesis}}_{\text{Q2'28}} \rightarrow \underbrace{\text{Submit}}_{\text{Q3'28}}$$